

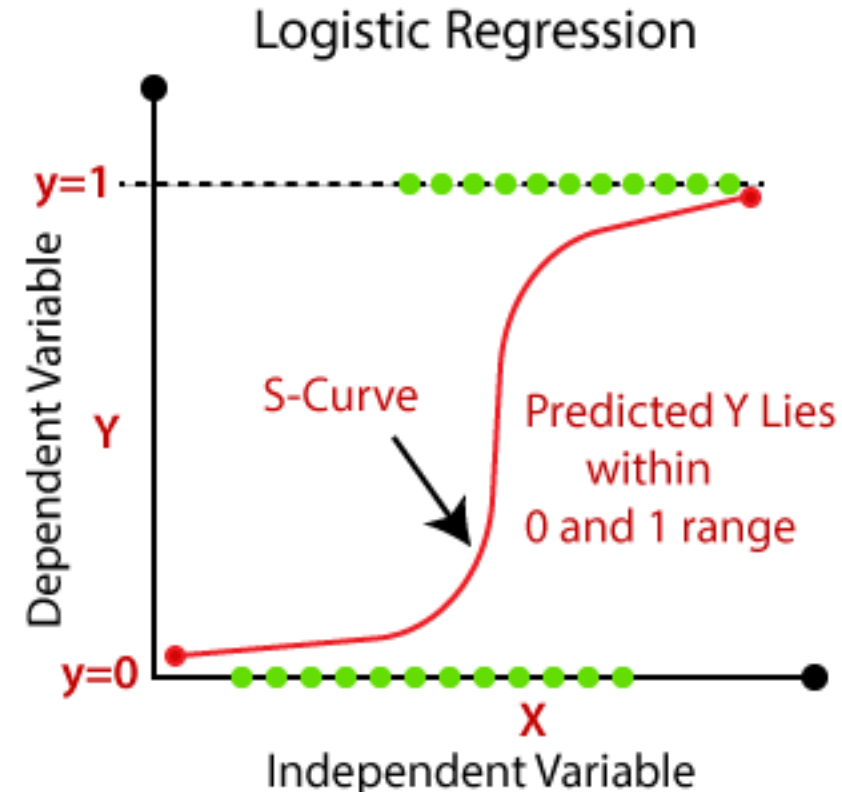
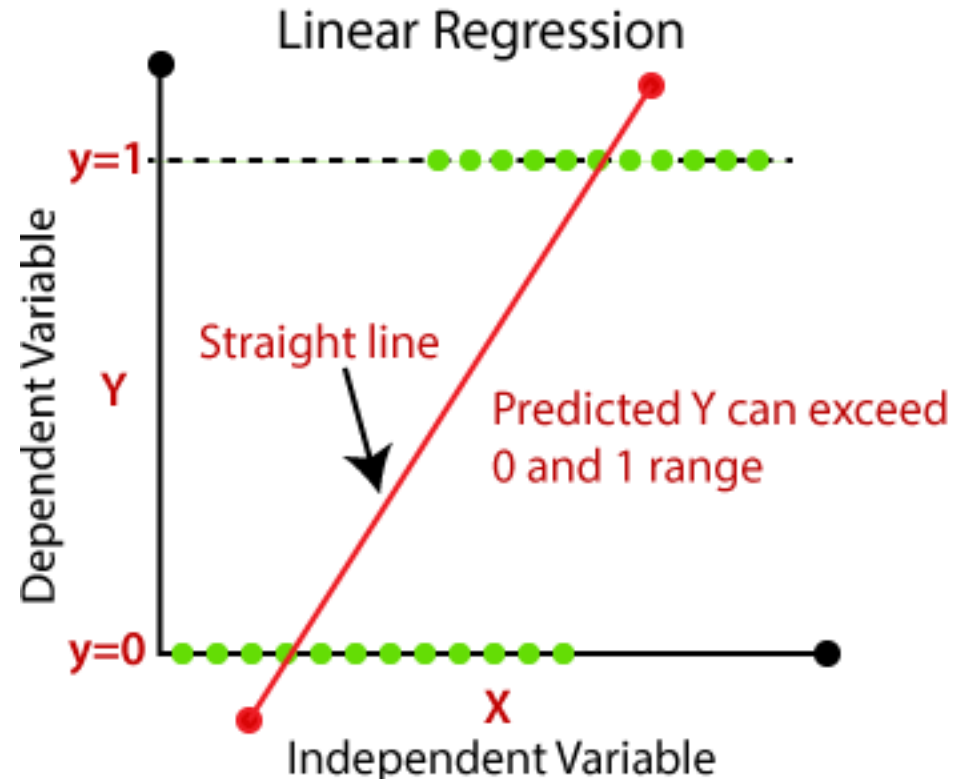
Logistic Regression

Logistic regression is a statistical method used for analyzing a dataset in which there are one or more independent variables that determine an outcome. The outcome is measured with a **dichotomous** variable (in which there are only **two** possible outcomes). It is used extensively in various fields, including machine learning, most often for binary classification problems.

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List of main areas where logistic regression is commonly used:

1. **Medical Research:** It is frequently used to identify risk factors for diseases and to predict the likelihood of a patient having a disease based on observed characteristics
2. **Financial Services:** In finance, logistic regression is used for credit scoring, predicting the probability that a borrower will default on a loan based on their financial history and other demographics.
3. **Marketing:** Customer behavior prediction like purchase likelihood.
4. **Social Sciences:** Analysing survey data to study binary outcomes like voting preferences.
5. **Manufacturing:** Predictive maintenance by forecasting machine failures.
6. **Education:** Predicting student outcomes such as exam performance.



Logistic Function (Sigmoid Function): The logistic function, also known as the sigmoid function, maps any real-valued number into the interval $[0,1]$, making it ideal for modeling probabilities. It is defined as:

$$f(x) = \frac{1}{1 + e^{-\beta x}}$$

Components of the Function:

- $f(x)$: This is the sigmoid function itself, mapping real numbers to the $(0, 1)$ interval.
- x : Represents the input to the function.
- β : A parameter that affects the steepness of the curve. The graph does not specify the value of β , so we typically assume $\beta = 1$ for simplicity unless specified otherwise.

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$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

Here's what each term represents:

- $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ are the coefficients (parameters) of the model. β_0 is the intercept, and $\beta_1, \beta_2, \dots, \beta_k$ are the coefficients of the predictor variables $X_1, X_2, \dots, X_k$.
- $X_1, X_2, \dots, X_k$ are the predictor variables (independent variables).

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Applications:

1. **Logistic Regression:** In logistic regression, the sigmoid function models the probability that the dependent variable belongs to a particular class.
2. **Neural Networks:** As an activation function, it helps determine whether neurons should be activated, facilitating non-linear decision-making across the network.

$$\log\left(\frac{y}{1-y}\right) = mx + c$$

1. Raising e to the power on both sides of the equation

$$\left(\frac{y}{1-y}\right) = e^{mx+c}$$

2. One divided by both sides of the equation

$$\left(\frac{1-y}{y}\right) = e^{-mx+c}$$

$$3. \left(\frac{1}{y} - 1\right) = e^{-mx+c}$$

$$4. \frac{1}{y} = 1 + e^{-mx+c}$$

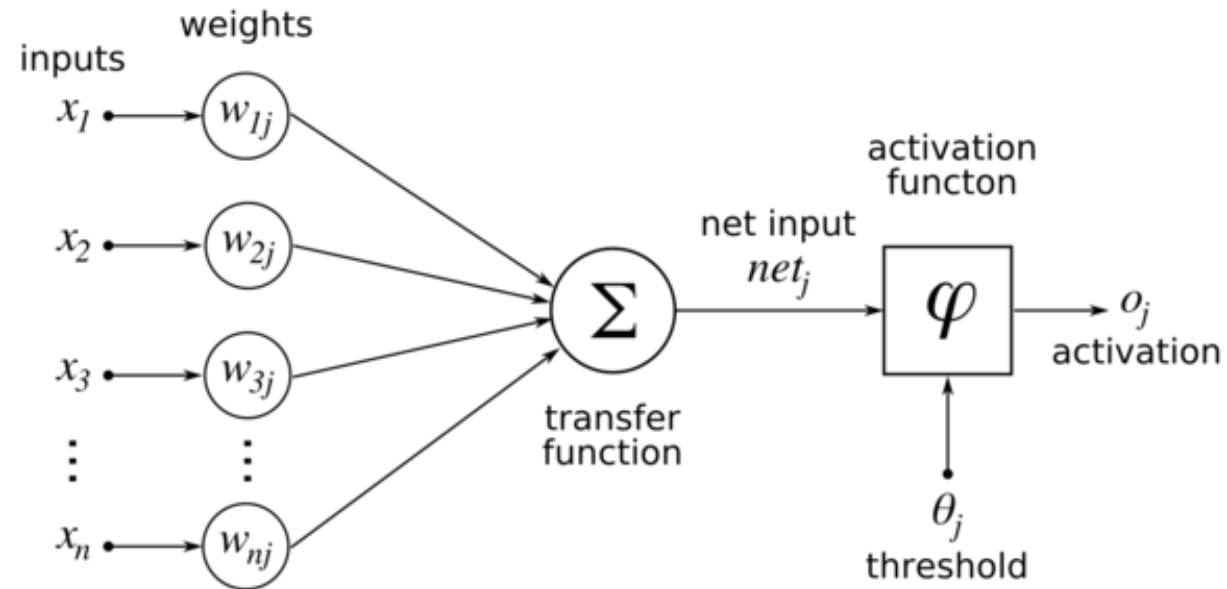
$$5. 1 = y(1 + e^{-mx+c})$$

$$6. y = \frac{1}{1+e^{-(mx+c)}}$$

Deep Learning Activation Function: The logistic function, commonly referred to as the sigmoid function, is frequently used as an activation function in neural networks, particularly in binary classification tasks. The use of the logistic function in this role stems from its ability to map input values (any real-valued number) to an output **range between 0 and 1**, making it ideal for tasks where the goal is to predict probabilities that are interpretable as such.

Why Use the Logistic Function as an Activation Function?

- ❖ **Output Range:** The logistic function outputs values in the range $[0,1]$ which is particularly useful for models where the output is interpreted as a probability. For binary classification, this output can be treated as the probability that a given input belongs to the positive class (usually represented by "1").
- ❖ **Non-linearity:** Neural networks require **non-linear activation** functions to help them **learn complex patterns** in data. The logistic function introduces the necessary non-linearity that allows the network to perform more than just simple linear transformations, thereby solving more complex decision boundaries between classes.



$$y_1 = \sigma(w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3 + b) = \sigma\left(\sum_i w_i \cdot x_i + b\right)$$

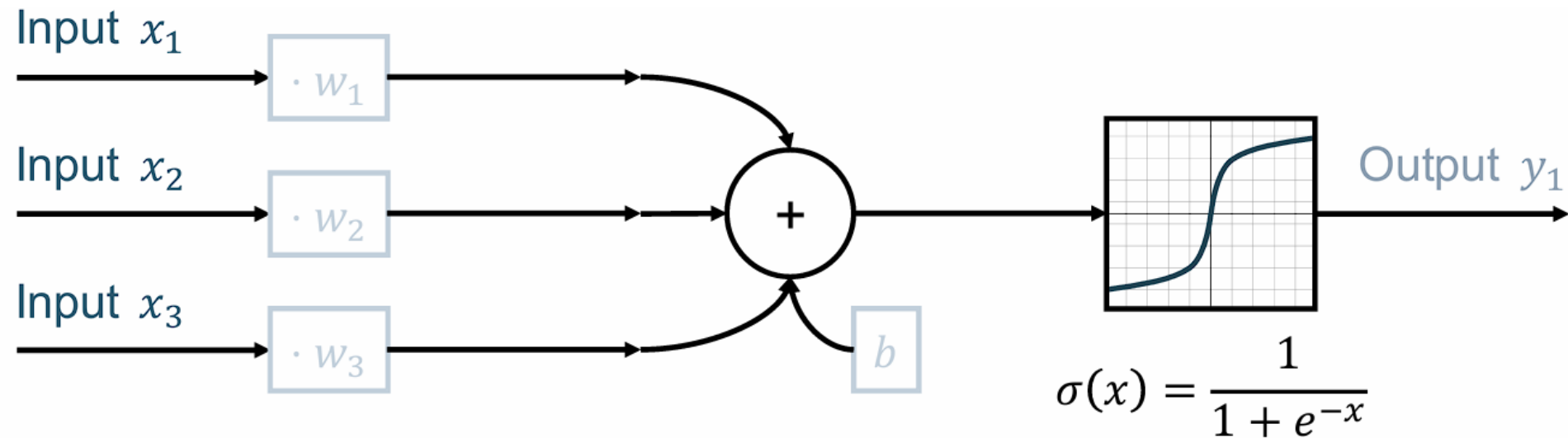
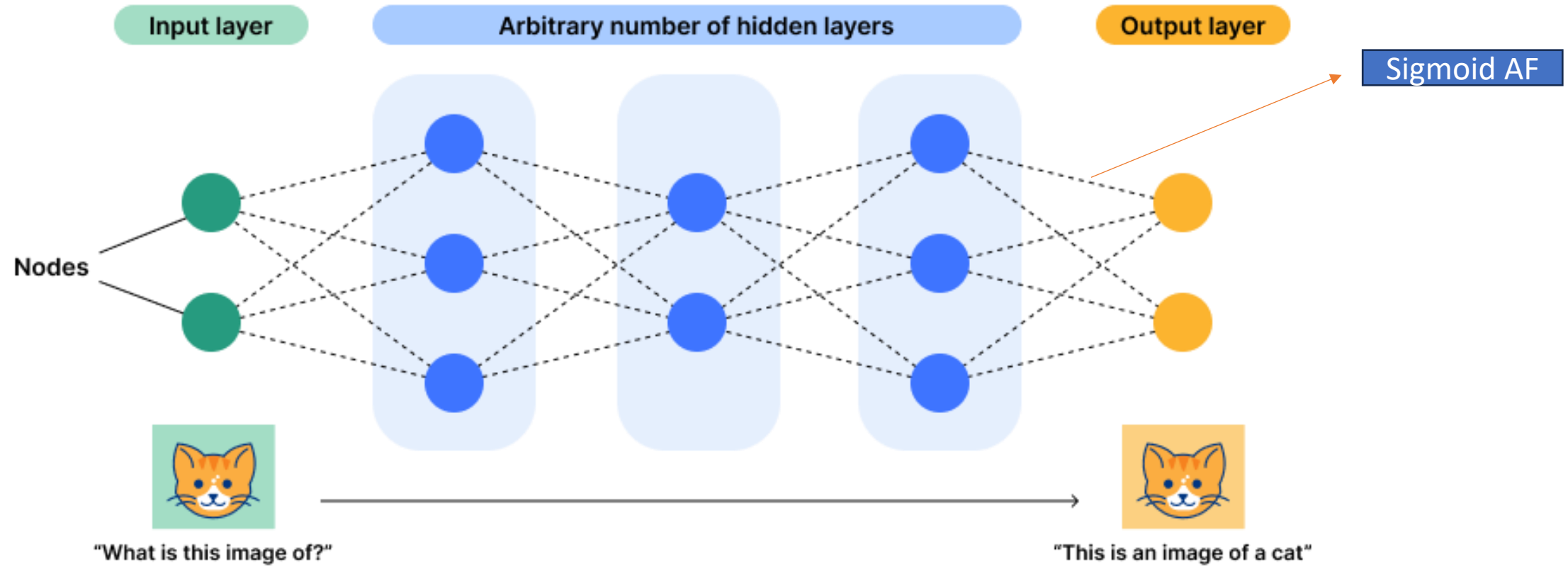


Fig: Artificial Neuron (Single Layer Perceptron)



Thank you